

Move Fast Without Breaking the Bank

Model Risk Management of GenAI workflows

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Executive Summary

The emergence of generative artificial intelligence (GenAI) in financial institutions promises major efficiency gains, while introducing novel and significant risks. This paper discusses considerations and mitigation strategies for managing model risks of GenAI, in light of supervisory expectations and industry best practice, both at an institutional level, and for particular techniques and specific uses of individual models.

GenAI systems are composable (For example, LLMs, retrieval indexes, prompt libraries, tool call orchestration, policy guards), dynamic (Such as, knowledge base updates, prompt revisions, API version drift), and open ended in their outputs. In addition, GenAI is envisioned to perform tasks in novel areas, and in particular, tasks currently performed by people. Specifying those tasks and managing the risk related to performing them automatically, requires substantial amounts of prompt and software engineering and poses challenges to how tasks are specified, qualified, quantified and controlled. Such properties expand the model risk surface and underscore the need to uplift model risk management approaches to account for the inherent complexity of GenAI and its potential uses.

Existing supervisory guidance, including US SR 11-7 and UK SS 1/23, establishes a clear framework of sound model risk management (MRM) principles, applicable to AI based models. Implementing these principles in the context of generative AI requires novel design, testing, and monitoring methodologies that advance in step with the ongoing development and complexity of these models, so that risk management remains aligned with technical progress and use of GenAI.

Given the wide area of potential application of GenAI in financial institutions, the large range of different technical implementations and designs, and the rapid change of both modeling techniques and potential applications, we have structured our coverage along three pillars **(i) Governance and tiering:** we discuss central issues that may arise when carrying out model identification, risk tiering, and change management **(ii) Design standards:** we identify the need to carefully identify, document, and justify GenAI model design decisions and emphasize common decisions that deserve attention **(iii) Testing and monitoring:** We explore how evaluation frameworks are broadening from static benchmarks toward task specific assessments, stress testing, and red-teaming approaches. This includes combining domain specific, general capability, and safety benchmarks, as well as employing adversarial probes and scenario based stress tests to capture rare failure modes. Monitoring business aligned risk metrics established during development enables mitigation with thresholds, escalation paths, and compensating controls.

Throughout our discussion, quantification of uncertainty emerges as a key challenge in GenAI, and improvements in this area could markedly improve risk management. Monitoring model uncertainty through calibrated confidence enables further risk mitigation including abstention, conformal style selective prediction, and conservative overrides with routing to humans or fallback systems embedded in well designed contingency plans.

In summary, managing model risk of AI is necessary for safeguarding the deployment of large, composable,

and evolving GenAI workflows at production scale. This requires careful consideration and integration of emerging design, testing and monitoring practices, and considerations of the novel failure modes and complications arising in the design, implementation and use of GenAI in a banking context.

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1 Introduction

Financial institutions have long relied on advanced quantitative models for tasks such as financial risk management, stress testing, and financial crime detection [1, 2, 3, 4]. The emergence of generative AI (GenAI) marks a shift toward wide scale use of applications that augment or replace human decision making and performance of tasks with potential widespread use and applications throughout all aspects of banking [5]. This expansion promises efficiency gains but also introduces novel and significant risks.

The use of novel modeling techniques, particularly when applied in unfamiliar domains, introduces the possibility of previously unrecognized risks— highlighting the strategic importance of robust model risk management (MRM). MRM provides a structured framework for identifying, assessing, and mitigating risks to ensure models are applied reliably and effectively.

Regulators have established frameworks to manage model risk. In the United States, the Federal Reserve's SR 11-7 guidance and related OCC guidance set out foundational principles for effective model risk management – encompassing robust model development, independent validation, and sound governance [6, 7]. Similar requirements have been put in place by other financial regulators, such as OFSE E-23 in Canada and SS1/23 in the UK [8]. SR 11-7 defines *model risk* as “the potential for adverse consequences from decisions based on incorrect or misused model outputs,” when viewed against the design objective and intended business use, which can manifest as financial loss, either directly via poor business decisions, or indirectly through fines or damage to an institution's reputation [6]. SR 11-7 defines a model as “a quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories to process input data into quantitative estimates” [6]; and provides the general framework that a model consists of three components – an input component (data and assumptions), a processing component (the calculation engine), and a reporting component that “translates the estimates into useful business information” [6].

Internal definitions of model risk and models, typically follow these relatively abstract definitions, to manage applications of complicated calculations for important business tasks, enshrined in overarching model risk policies and underlying validation guidelines and model development standards for specific model areas and has been the basis for model risk management for decades.

However, the deployment of modern AI systems – especially large language models (LLMs) and composable AI pipelines – presents several challenges to how model risk is managed, for developers and users, and model validators. For example, the boundary between data risk and model risk can potentially be blurred and the ability to produce unstructured outputs (such as freeform text or API's tools calling) means that methodologies for managing model risk must evolve in step with the technology. Regulators are beginning to acknowledge this shift – for example, recent supervisory updates in the UK explicitly broaden the scope of model risk management to encompass AI/ML models and even dynamic algorithmic tools that adapt over time [8, 9]. A comprehensive model inventory, sound governance, robust development and documentation, ongoing monitoring, and independent validation, continue to provide the foundation for use of Generative AI (GenAI) and composable AI workflows in financial services, but the complexity, dynamism, and scale necessitate updated methodologies that can operationalize these principles effectively.

To illustrate why that is the case, consider several prevalent AI deployment patterns now entering banking workflows:

Retrieval-Augmented Generation (RAG). An LLM is combined with a retrieval mechanism that fetches information from a knowledge base to ground responses in context. The model's output in a RAG system includes both a generated free text answer and references to source documents. Crucially, performance depends not only on the LLM's parameters but also on nonmodel artifacts like the embedding database, search index, and prompt templates that guide retrieval [10]. This means that a RAG's behavior can change substantially with updates to the underlying document repository or prompt logic, even if the core LLM is unchanged.

Fine-tuned and Domain Specialized LLMs. Financial firms are increasingly taking general purpose foundation models and adapting them to enterprise data (For example, fine-tuning on proprietary text from regulatory filings, loan documents, or internal policies). These fine-tuned models aim to improve accuracy

in specific domains and reduce undesired behaviors. However, even relatively small fine-tuning adjustments can materially shift behavior, blurring the line between a “new” model versus a mere variant of the base system [11]. The risk is compounded by vendor dependence: few banks can train large models from scratch, leaving them reliant on proprietary platforms where architectures, datasets, and training methods are undisclosed. Banks remain accountable for vendor acquired models under SR 11-7, yet transparency is limited [12].

Agentic Workflows. Here, an LLM is used not just to answer a single query, but to autonomously plan and execute a sequence of tasks – for example, an AI agent that reads an email, calls an API, runs a calculation, and then drafts a report based on results. Such agentic workflows (often implemented by chaining multiple LLMs and tools, in conjunction with classical machine learning techniques) produce dynamic action sequences rather than a single prediction or score [13]. The orchestration logic, prompt chains, tool calling frameworks or memory modules, all directly affects outcomes. Such models are especially challenging: developing agentic AI workflows takes not only a high degree of prompt engineering, but a significant effort of general software development and engineering, to specify inputs, outputs, intended uses and appropriate tests and controls. Additionally, the “output” to be tested can be an entire workflow, and intermediate reasoning steps are opaque. These models can exhibit emergent behaviors – compounding errors, unexpected tool usage, or novel strategies – that require sophisticated testing [14].

A complicating factor is that such models are rarely used in isolation. This composability accelerates innovation but complicates model risk management. While AI pipelines can be assembled quickly, verifying their safety, security, robustness, and compliance take far longer [15].

Recent industry research and supervisory commentary highlight several considerations to ensure robust AI model risk management [9, 12].

First, there is the tension between static vs. dynamic model assumptions. AI systems can evolve continuously without formal redevelopment— for instance, updating a RAG knowledge database, changing an LLM’s prompt templates, or an API provider adjusting an LLM’s behavior can all significantly alter outputs on the fly. Such changes call for more continuous monitoring and configuration management [16, 14]. Second, data preprocessors, feature embeddings, prompt libraries, and orchestration logic all of which effectively become model parameters by shaping outcomes, and thus demand governance and version control [15, 8]. Third, the lack of transparency in reasoning within advanced AI poses issues as it is often difficult to observe or explain the intermediate reasoning that leads to an answer or action [17]. Finally, emergent behaviors and novel failure modes demand expanded outcome testing. Generative AI is prone to issues with plausible but false outputs and unpredictable responses to rare inputs [18]. These are failure modes that require additional controls and testing, employing methodologies explicitly designed to evolve in step with advances in AI technology [8, 12]. Likewise, models used for newly developing areas may lack historical data to test against, resulting in more reliance on stress scenarios and qualitative analysis than on statistical backtesting [19].

Appropriate and proportional model risk management is pivotal for use of AI in banks. We recognize, however, that AI use also implicates a wider constellation of risks * Such risk categories and their controls are typically mapped to the enterprise risk taxonomy and overall AI risk appetite, consistent with supervisory expectations. Coordination among control functions is essential because GenAI risk is use case dependent and cross-cutting, and central standards are required to ensure consistency, proportionality and timely escalation as adoption scales and supervisory expectations evolve.

In the following, we touch upon a few associated risks and related areas but have otherwise kept the focus on model risk management of AI based models.

*Including (non-exhaustively) operational and ICT risk, information security, legal and compliance, conduct, strategic and project risk, data governance and privacy, third-party and concentration risk, reputational risk, access and identity management, intellectual property, resilience/availability, and health and safety.

2 Model Risk Classification and Tiering

Effective model risk management of individual models starts with identification of model risk at inception, resultant identification and registration in the model inventory and initial classification of model risk and tiering, typically involving assessments of model complexity (of technical solution and task to be performed), model exposure (materiality of potential financial loss) and model reliance (possibility to detect model misperformance, override outputs, measure uncertainty or availability of fallback or contingency plans). Identification of AI based models are complicated by the potential widespread use, and in business areas that have not traditionally been accustomed to use of models and resulting model risk management. Model risk tiering of GenAI is complicated by the complexity of the tasks and solutions, the often indirect financial impact and different aspects of model reliance.

Here, we give a brief outline of the general governance setup related to models, some challenges related to model identification and tiering and well as general change management of GenAI based models.

2.1 General Model Governance

The general principles for model risk management within banks are provided in Model Risk Management Policies, in line with regulation such as SR 11-7 and UK SS 1/23. This includes definitions (especially of model risk and models), general principles of model governance, including roles and responsibilities such as 1LoD model owners, developers, and users as well as for 2LoD model validators and MRM functions.

First line (Business) 1LoD owns business cases and use cases, articulates business intent and are responsible for the development (including potential procurement), testing, documentation, use and monitoring of models. They are responsible for identifying models and submitting them to model validation prior to deployment. In addition, they are responsible for obtaining any IT approval or other types of approval related to models for example from an IT application perspective, legal approvals, and product approvals.

Second line (Model risk and validation functions) 2LoD model risk and model validation functions are responsible for independent and effective challenge of use of models and management of model risk by model owners, users and developers. Validation activities are typically carried out by dedicated teams responsible for specific model areas. These include regulatory “Internal Models” for Credit Risk, Counterparty Credit Risk, Market Risk, and Pricing, as well as other business critical models such as those used in treasury and asset–liability management (ALM), fraud detection, and financial crime prevention.

First line of defense In the context of GenAI, development teams are responsible for the end-to-end design and documentation of models. This includes: (1) selecting architectures and data sources; (2) defining intended use and business objectives; (3) specifying inputs and outputs; (4) documenting design choices such as prompt templates, retrieval configurations, or tool integrations; (5) implementing robust testing (For example via sensitivity analysis, stress testing, benchmark evaluation) and recording limitations; and (7) designing monitoring metrics and fallback strategies for deployment. Each of these steps must be comprehensively documented. For GenAI models, which rely heavily on heuristic design and vendor APIs, the development team also maintain a *dependency register* of external tools, prompts, and RAG knowledge bases. Developers are expected to demonstrate conceptual soundness and alignment with business purpose.

Second line of defense (model risk and validation). Validation teams are charged with independently challenging the assumptions, methods, and evidence provided by developers. Independence is not only achieved by organizational separation, but also by not *de facto* acting as model developers. This typically puts limitations on how much specific guidance on model development that model validation units can provide.

In practice, the validation team: (1) assess whether the model’s purpose and design are consistent, and that models are sufficiently documented, and in line with regulatory and policy expectations; (2) replicate or challenge testing conducted by the developers, using alternative datasets, adversarial perturbations, or different evaluation frameworks; (3) probe the robustness of GenAI specific artefacts such as prompt engineering choices, retrieval pipelines, and tool connectors; and (4) assess whether models are implemented

and used as specified; (5) assess adequacy of data and model monitoring; (6) provide a clear validation opinion that addresses conceptual soundness, empirical performance, and limitations. Validators additionally assess whether monitoring plans, fallback mechanisms, and uncertainty quantification procedures are adequate to detect and mitigate risks in live operation.

Model validations are performed prior to deployment of models (initial validations) and on an ongoing basis (ongoing validations/model reviews). Ongoing validations are triggered either by events (For example, changes to the model, to its behavior, model use or model environment), or by passing of time.

The primary outcome of the validation activities are model validation reports, that document the activities performed by the validators as part of the model validation activity. These documents form the basis for decisions related to approved use of model output for specific business tasks, and serve as documentation of the independent and effective challenge by 2LoD of model use and model risk management in 1LoD. The model validation reports typically contain descriptions of the model(s), documentation, software and data (input, outputs, parameters, monitoring), as well as descriptions, assessments and tests performed by the model validators as part of the effective challenge of model use and model risk management. The model validation reports typically result in findings to be remediated by model owners, developers or users, and an overall conclusion on whether models are fit for purpose and can be recommended for use. Naturally, the specific validation activities depend on the modeling techniques and the specific intended and actual use of model output, and is proportional to the degree of inherent and residual model risk present.

In addition to responsibility for model validations, the 2LoD model risk and validation functions also are responsible for overarching model risk policies, model validation guidelines, maintaining a model inventory, and reporting on status of models and validations at an aggregated level. In principle, the activities performed by model validators as part of model validations are not necessarily different than what could be performed by 1LoD model owners, developers and users as part of model development, use and monitoring.

2.2 Model Identification and Risk Tiering

Identifying *where* GenAI models and their outputs are used is a central operational challenge: without a reliable discovery process, critical use of models, and dependencies will be missed. Some challenges arise in terms of applying the model definition, and delineating, for example, what constitutes models, submodels, or feeder models, that are also present for legacy models, but will likely pose additional challenges for GenAI, especially for more complicated setups, workflows and agentic constructions.

Generally, we recommend registering the *full GenAI workflow* as the primary governed object, with the large language model(s) (LLM's), prompt templates, retrieval-augmented generation (RAG) components, orchestration code, and tool connectors recorded as explicitly governed subcomponents. This aligns with the SR 11–7 expectation that inputs, processing, and reporting be documented and controlled, while acknowledging that behavior in GenAI systems is materially shaped by nonmodel artefacts and configuration choices [6]. In practice, institutions should (i) instrument usage telemetry across products and business processes to surface undisclosed model touchpoints, (ii) maintain a live dependency bill-of-materials for external tools/APIs with versioning, scopes, and rollback plans, and (iii) treat *material* tool or configuration changes as model changes for purposes of approval, revalidation, and change control [8].

Traditional tiering frameworks in banking typically rely on two complementary approaches. **Decision tree tiering** encodes a small set of high signal criteria (For example, materiality, customer/regulatory exposure) into an interpretable flow that assigns models to low/medium/high tiers; its strengths are transparency and consistency, but it can miss nuance in multi-factor settings. **Scorecard tiering** aggregates weighted factors (such as, reliance, complexity, uncertainty, visibility) into a composite score mapped to tiers; it captures nuance and factor trade-offs but needs careful calibration and governance. In practice, firms often add an **expert overlay/committee review** to adjust borderline cases and ensure alignment with business context and enterprise risk appetite.

GenAI and agentic workflows require targeted augmentations to these approaches as composable architectures (LLM + RAG + tools) are dynamic and heavily influenced by nonmodel artefacts. We

propose also incorporating the following GenAI specific drivers into either decision tree gates or scorecard factors: *autonomy* (ability to act or chain tools without human approval), *content sensitivity* (regulated or confidential data), *nonmodel dependencies* (retrievers, prompt libraries, orchestration/toolchains) and their *change velocity*, and *evaluation uncertainty* (such as robustness of output, robustness to prompt variation, reasoning reliability). These drivers are motivated by documented properties of foundation models and their evaluation challenges [11, 15], dynamic behavior introduced by retrieval and configuration changes [10, 14], and new failure modes in agentic settings (tool-call errors, compounding plans, security risks) [18]. Concretely, high-autonomy agents or external-facing RAG systems with volatile corpora should default to higher tiers (triggering deeper validation/monitoring), whereas tightly guarded internal advisory uses with stable artifacts may sit lower, subject to guardrails and continuous oversight. The aim would not be to replace existing tiering, but to extend it with factors that reflect GenAI's system-level risk profile and the realities of composable AI, in light of the particular technical solution and the specific use of input and output.

Effective tiering therefore, requires assessment of materiality, potential customer impact, and regulatory sensitivity, as well as more hidden risks such as RAG corpus integrity, and orchestration fragility. Institutions should codify joint review panels or overlay committees to balance these perspectives, with tier outcomes driving testing and documentation levels, monitoring cadence, as well as validation depth, in line with SR 11–7 and SS1/23 [6, 8].

2.3 Change Management

Material changes to model parameters, APIs, prompt/orchestration logic, retrieval corpora, or guardrails should be notified in advance and treated as in-scope for change control; institutions should retain the artefacts needed to perform impact assessment and revalidation under ongoing monitoring.

While reliance on vendor models does not, in itself, create novel *types* of validation challenges compared with other third-party models, GenAI supply chains (closed models, fast API iteration, RAG indices, toolchains) heighten the need for secure-by-default designs and continuous, *evidence-based* assurance [20, 21]. This direction of travel aligns with emerging operational resilience obligations for third-party ICT providers in the EU (DORA) and UK oversight of critical third parties (CTPs), which push for greater transparency, incident reporting, and testing of external dependencies [22, 23, 24]. Practically, financial institutions should codify vendor deliverables (transparency reports, evaluation traces, red-team results), pre-pre-deployment security artefacts, and advance-notice change SLAs into contracts, and integrate vendor updates into the bank's model inventory, tiering, and monitoring cycle.

Vendor management for GenAI and agentic workflows should extend established third-party risk practices, with added emphasis on transparency, change control, and software supply-chain security. At minimum, vendors should furnish durable documentation (akin to *Model Cards* and *Datasheets for Datasets*) that summarizes methodology, high-level architecture, intended use, limitations, and key assumptions, alongside evidence from multi-metric evaluations (For example, groundedness for RAG, tool-use accuracy for agents) [25, 26, 15, 27].

3 Model Design

Model development, including rationale for model design choices, must adhere to rigorous standards of conceptual soundness and reliability, as mandated by supervisory guidance. Both U.S. and U.K. regulators emphasize that model development should be a *disciplined process* with clear objectives, sound design, robust methods, and strong documentation. SR 11-7 states that:

A sound development process includes: a clear statement of purpose; sound design, theory, and logic; robust model methodologies; rigorous assessment of data quality; and appropriate documentation (...) and that testing is an integral part of model development... to show the model is performing as intended; to demonstrate that it is accurate, robust, and stable; and to evaluate its limitations and assumptions.

Similarly, the U.K. PRA's SS1/23 requires firms to maintain

a robust model development process with standards for model design and implementation, model selection, and model performance measurement, (...) with regular testing of data, model construct, assumptions, and model outcomes.

Though principled mathematical theories drive many of the fundamental design advances in machine learning and artificial intelligence, today's state-of-the-art GenAI systems demand thousands of nuanced engineering choices – choices that must be made in line with regulatory standards. Increasingly, these choices lack principled mathematical rationales for their selection and instead emerge as *de facto* defaults—heuristics adopted from practice rather than theory [28]. Examples include prompt template design, reinforcement learning from human feedback (RLHF) strategies, data filtering heuristics, and architectural tuning rules. Such design choices materially affect outcomes, yet they often lack the transparent theoretical grounding that regulators traditionally expect. As SR 11-7 states, even models with significant qualitative elements must be developed and justified in a sound manner [6].

This applies to LLM's in isolation, as well as when LLM's or other types of GenAI models are used as submodels or components in more complicated designs, such as agentic workflows, and in light of the particular use of such models and potential actions taken by them.

Every modeling choice—no matter how heuristic—requires explicit documentation, justification, testing, and independent challenge. In addition, *model development relies heavily on the experience and judgment of developers* [6], so organizations should ensure that the development team has appropriate expertise (For example, in NLP, machine learning, and domain knowledge). Engineering governance practices such as version control, structured success criteria for development phases, and early involvement of risk teams help ensure such choices are traceable and auditable. Notably **clear documentation** is paramount; SR 11-7 and SS1/23 both insist that development documentation be detailed enough that an independent party can understand how the model works and on what assumptions, and even replicate the outcomes [6, 8]. This includes describing data sources, data preprocessing or filtering steps, the modeling methodology (such as, architecture, training procedure, hyperparameters), and any “expert judgment or conservatism applied to address model limitations” [8].

During development, rigorous model testing, to affirm development decisions and final model choices are well founded, is critical. SR 11-7 highlights testing as essential to demonstrate that a model is “*accurate, robust, and stable*” [6], while SS1/23 calls for performance evaluation across diverse conditions, including stress and sensitivity tests to establish operating boundaries [8]. For GenAI, robustness requires testing not only on training-like data but choosing models by considering unusual or potential adversarial inputs, where phrasing or context shifts can trigger unexpected failures.

Another fundamental principle is alignment with **intended use**. Developers must clearly define the purpose and scope of the GenAI model from the outset and ensure that all development decisions align with that purpose [8]. A key part of intended use is limiting model failure by establishing principled AI monitoring plan during development, for both inputs, outputs and internal parameters. SS1/23 emphasizes that firms

should define performance metrics, monitoring thresholds, and escalation triggers before deployment [8]. This ensures a smooth transition into production and allows early detection of drift or emergent risks.

Finally, firms should document fallback strategies and contingency plans for revoked use of models. In practice, this may mean routing low-confidence outputs to human review, reverting to simpler challenger models, or deploying deterministic rule-based systems as backups. Together, robustness testing, monitoring plans, and fallback mechanisms form the foundation of development practices that make GenAI systems ready and resilient.

In the remainder of this section, we focus on five areas where uplift of standard model risk practices is most needed: (i) careful metric selection, (ii) learning algorithm design, (iii) sensitivity analysis, (iv) data quality assessment, and (v) explicit uncertainty quantification. These focal points directly reflect the supervisory emphasis in SR 11-7 and SS1/23 on robust testing, data governance, and sound methodological design [6, 8]. Addressing them systematically is essential for ensuring that GenAI systems deployed in financial services are not only performant and resilient, but also validation ready, and aligned with regulatory expectations.

3.1 Establishing Fitness Metrics

A central component of GenAI model development is the definition of *fitness metrics*—the criteria by which a model's performance and suitability are evaluated. Fitness metrics and loss functions define the signals to be optimized, thereby guiding the selection of a model most fit for purpose. Misalignment of these functions with business use cases will result in foundational model errors and significant model risk. Metrics apply both on the totality of the model, in light of its intended use, for example, in agentic workflows, or to model components, such as LLM's in isolation.

Recent research categorizes GenAI evaluation metrics into three groups: **reference-based**, **reference-free**, and **LLM-based** metrics [29, 30]. Reference-based metrics, such as BLEU, ROUGE, and BERTScore, compare outputs against a ground-truth reference, providing concrete measures of overlap and semantic similarity [31, 32]. Reference-free metrics, such as SUPERT or BLANC, assess the quality of generated outputs without explicit references, focusing on dimensions like informativeness, fluency, or logical entailment [33, 34]. More recently, LLM-based metrics (sometimes called “LM-as-a-judge” approaches) have emerged, where a GenAI model is used to rate outputs on attributes such as coherence, truthfulness, or usefulness. Frameworks such as HELM illustrate that a combination of metrics might be necessary, with evaluations spanning multiple axes including accuracy, calibration, robustness, and fairness [35].

For financial applications, additional domain-specific fitness metrics are required. Examples include compliance rates with consumer protection rules, factual accuracy of generated research summaries, or plausibility of scenario narratives for stress testing. Supervisory expectations are explicit: SR 11-7 and SS1/23 require that performance tests be conducted across a range of plausible scenarios, with metrics aligned to the model's intended business use [6, 8]. In practice, institutions should document why chosen metrics are appropriate proxies for success in context, and how they reflect the model's design objectives

3.1.1 Loss Functions and Risk Metrics

Scope and misalignment. In GenAI workflows, the training *loss function* (what is optimized) rarely coincides with the *risk metrics* that matter for governance and use. Large language models (LLMs) are typically trained with next-token likelihood (cross-entropy) and subsequently aligned on proxy rewards via preference learning (For example, RLHF, DPO), which typically target fluency or generic helpfulness rather than domain-specific truthfulness, compliance, or safety [36, 37]. Consequently, development and validation efforts must explicitly evaluate design decisions with *risk-facing* metrics that reflect the application's goals and constraints rather than inferring assurance from training loss alone. Such evaluations will be the focus of the rest of this section.

Reference-based metrics. When ground truth outputs exist, overlap and semantic similarity measures provide reproducible signals. Classical n-gram metrics (BLEU, ROUGE) and embedding-based measures

(BERTScore) remain useful baselines for text generation and summarization [31, 38, 32]. For factual consistency in summarization and longform answers, NLI-style evaluators such as FactCC and SummaC measure contradiction and entailment errors against sources [39, 34]. In retrieval-augmented generation (RAG), attribution- and source-grounded metrics (For example, RAGAS, AttributionBench, RAGTruth) assess whether answers are supported by retrieved passages and quantify hallucination or unsupported claims in a document-grounded setting [40, ?, 41].

Reference-based metrics provide an interpretable and reproducible starting point for assessing GenAI systems, particularly when historical or human-generated outputs exist for comparison. However, their applicability is constrained by the inherent multiplicity of valid outputs in generative tasks, the potential for bias or error in reference datasets, and the risk of model overfitting to narrow lexical overlap criteria [42, 43]. Consequently, while such metrics are useful for benchmarking and regression testing, they cannot be regarded as comprehensive indicators of reliability or robustness.

Reference-free metrics. When unique references do not exist, reference-free indicators capture intrinsic qualities and distributional fit. Perplexity and entropy capture fluency and uncertainty; distributional divergences and diversity measures monitor mode collapse; and tone of voice/toxicity and bias detectors (For example, RealToxicityPrompts) provide indicators of inappropriate content [44]. For longform factuality, zero-reference detectors such as SelfCheckGPT and entropy-based uncertainty estimators in black-box LLMs detect a subset of inaccurate output with measurable statistical behavior [45, 46]. These signals are well suited for *production monitoring* because they can be computed at inference time without labels.

However, such metrics often fail to align perfectly with human judgments: measures like perplexity capture surface fluency but may assign high scores to outputs that are semantically incoherent or factually incorrect [47]. Moreover, many reference-free approaches rely on auxiliary models—for example, natural language inference (NLI) classifiers to assess consistency—meaning that their validity is contingent on the accuracy and robustness of those secondary systems [48]. This introduces the risk of error propagation, where weaknesses in the auxiliary evaluator distort the assessment of the primary model.

LLM-as-judge (LMJ) metrics and guarantees. LLM-based evaluators (“LM-as-a-judge”) scale qualitative assessment by prompting a stronger model to rate or rank outputs [30, 49]. While LMJ approaches correlate with human judgements on many language tasks, they exhibit systematic brittleness (Such as, length/style bias, prompt sensitivity, distribution shift) [50, 51]. Recent work proposes calibration and procedural safeguards (pairwise/tournament protocols, rubric-guided prompting, bootstrap uncertainty) to mitigate bias and to report confidence bands rather than point scores [49]. For regulated use, LMJ pipelines should be (i) *frozen and versioned*, (ii) *calibrated* to human panels on task-specific samples, and (iii) supplemented with *statistical uncertainty* reporting to avoid compounding evaluator risk.

Reasoning fidelity and logic monitoring. For chain-of-thought and tool-using agents, correctness depends on *procedural reliability*, not just final answers. Self-consistency aggregates multiple reasoning paths to reduce stochastic error [52], while faithfulness diagnostics test whether intermediate rationales are causally tied to correct predictions rather than post-hoc justifications [53]. Verifier paradigms (such as, “verify-then-edit” and step-level proof checks) further separate *solution finding* from *solution checking*, improving robustness under distribution shift [54]. These techniques motivate *logic monitors* in production: trace logging of plans, step-level checks, and disagreement detectors.

Tool use, orchestration, and attribution of decisions. Agentic LLMs increasingly *plan* and *call tools* (search, retrieval, calculators, APIs) [55, 56, 57, 58]. Risk metrics should therefore include (i) *tool call justification rates* (was the call necessary and beneficial?), (ii) *attribution fidelity* (are claims grounded in tool outputs?), and (iii) *rollback safety* (does the system fail safe on tool errors/timeouts?). In RAG and API-augmented settings, evaluations must treat embeddings, indices, prompts, and routers as *governed components* with versioned benchmarks [40].

Static vs. dynamic benchmarks. System-level suites such as HELM provide multi-axis evaluation across accuracy, robustness, calibration, fairness, toxicity, and efficiency, establishing comparable baselines [15]. However, static leaderboards are insufficient for financial applications with shifting distributions, bespoke policies, and evolving tools. Dynamic, task-specific evaluations (adversarial data collection, on-policy replay, and scenario-based probes in the spirit of Dynabench) are required to surface failure modes that

static sets miss and to prevent overfitting to benchmarks [59].

Market-facing risks and emergent behavior. In economic decision settings, multi-agent learning can exhibit tacit collusion and other emergent behaviors even without explicit communication, underscoring the need for *system-level* outcome metrics (For example, price dynamics, welfare impacts) [60, 61]. Financial GenAI evaluations should therefore include stress tests and guardrails for strategic interaction, not only single-agent accuracy.

Training losses for parameter optimization and evaluations metrics that guide model design choices are critical choices to justify during development and validation. A defensible GenAI evaluation stack combines (i) reference-based factuality/attribution where ground truth exists, (ii) reference-free quality and safety monitors for live traffic, (iii) calibrated LM-as-judge protocols with uncertainty reporting, and (iv) reasoning, tool use, and system-interaction checks. Static suites establish baselines; dynamic, task-specific probes keep pace with evolving data, prompts, and tools.

3.2 Sensitivity Analysis

Sensitivity analysis evaluates how small, controlled variations in input or context influence a GenAI model's output—a critical discipline for ensuring robustness in financial applications. Regulatory guidance (SS1/23) mandates sensitivity testing to establish performance boundaries, and analogous stress-testing is central to SR 11-7's principle of conceptual soundness [8, 6].

Prompt sensitivity. Recent research methods elevate traditional perturbation approaches with systematic frameworks. For instance, the Latent Adversarial Paraphrasing (LAP) technique employs a dual-loop adversarial optimization to identify worst-case semantically equivalent paraphrases that can drastically affect model behavior, enabling targeted robustness improvements [62]. Similarly, SelfPrompt autonomously generates domain-constrained adversarial prompts via knowledge-graph guided perturbations, offering efficient robustness evaluation tailored to operational domains—without relying on static benchmarks [63].

Pipeline-level sensitivity. Robust GenAI workflows require *systemic* sensitivity testing that perturbs not only prompts but also the surrounding pipeline. For retrieval-augmented systems, developers should ablate or degrade the *index* (For example, staleness, partial corruption, embedding swaps) and measure degradation in grounding/faithfulness, since RAG behavior can shift materially under retrieval noise or targeted poisoning [?, 64, 65]. Concretely, we recommend (i) simulating document additions/deletions at various freshness cadences, (ii) injecting adversarial distractors at controlled rates, and (iii) swapping retrievers/embedding models to quantify cross-component brittleness. Sensitivity curves should be reported for factual consistency (such as NLI-based entailment of answers by retrieved context) and answerability under index faults [66, 40].

Loss-objective and training-protocol sensitivity. Outcome risk often depends on alignment/training choices that are *not* equivalent in their failure profiles. We therefore propose controlled sensitivity to alternative objectives and protocols—for example, SFT-only vs. RLHF [67], DPO [37], KTO [68], and “Constitutional” AI feedback [69]. Institutions should quantify how hallucination, toxicity, and distributional robustness change across these objectives and hyperparameter sweeps (preference weight, temperature/top- p , KL penalties) and document any Pareto frontiers observed between helpfulness and safety metrics. These experiments provide *design justification* and reduce hidden objective-induced risk.

Tool-use dependencies and access-control sensitivity. Agentic workflows introduce *tool-call* dependencies whose reliability and security should be stress-tested. We recommend whitelisting tools and evaluating API selection and schema adherence under controlled perturbations: degraded availability (timeouts), version/API schema drift, permission scope changes, and ambiguous tool lists. Empirical work on tool learning (For example, Toolformer [55], Gorilla [58], ToolLLM [56]) shows that tool routing is sensitive to prompt context and API descriptions; accordingly, monitoring should log (and justify) each tool call with input/output schemas and confidence and classify *material* tool changes as model changes. Security sensitivity should include prompt injection and data exfiltration tests at tool boundaries [64].

Reasoning-logic evaluation and process robustness. For chain-of-thought or multi-step planners, sensitivity should target the *process*, not only final answers. Recent work on process supervision and verifiers (PRM/step-level reward models, verifier models) demonstrate that monitoring intermediate steps detects latent reasoning faults earlier than outcome-only checks [70, 71, 72]. We propose (i) randomized reordering or occlusion of intermediate steps, (ii) “self-consistency” resampling with semantic agreement scoring, and (iii) verifier- or rule-based checks on step invariants for finance tasks (For example, cash-flow identity, P&L additivity). These techniques should be complemented by automated hallucination/self-checking detectors [45].

Multi-agent and market-interaction risks. Where agents interact (such as simulated markets, pricing, or negotiation), sensitivity should include *strategic* behaviors. Decades of literature show that adaptive algorithms can exhibit tacit collusion without explicit communication [60, 61]. Early evidence indicates that learning agents can coordinate in repeated games under plausible reward designs [73]. Thus, agentic sensitivity should sweep opponent pools, reward shaping, and disclosure policies, and include explicit anti-collusion tests (price dispersion, response to unilateral deviation, and stability under perturbations).

Outputs: operating envelope and fallbacks. Finally, modelers should include scenario-based robustness checks—such as extreme market crashes or ambiguous customer messages—and systematically vary generation parameters (For example, temperature, context length) to assess drift and stress points. The outputs define a model’s operational envelope: quantifying ranges in which reliability and compliance hold, and flagging regions that require guardrails or human oversight. All pipeline/logic sensitivities should culminate in a documented *operating envelope*: validated input regimes, index freshness bounds, permissible tools/versions, and process invariants. Production controls should include a monitoring plan (faithfulness, toxicity/bias, tool-call anomaly rates) and *fallback policies* (e.g., human-in-the-loop or conservative deterministic baselines) triggered by threshold breaches [6, 8].

General approaches to sensitivity analysis remain valid and sound; however, understanding where and how such techniques should be used within a GenAI model is critical to appropriate model selection and avoiding miscalibrated risk estimates.

3.3 Assessment of Data Quality and Relevance

Establishing clear, well-documented justification for the quality and selection of data used during model development is critical. SR 11–7 states (emphasis added):

The data and other information used to develop a model are of critical importance; there should be rigorous assessment of data quality and relevance, and appropriate documentation. Developers should be able to demonstrate that such data and information are suitable for the model and that they are consistent with the theory behind the approach and with the chosen methodology. [...] This is particularly important for external data and information (from a vendor or outside party), especially as they relate to new products, instruments, or activities.

This subsection addresses *development* data (training and adaptation corpora), as distinct from *evaluation* datasets and benchmarks covered in Section 4. In practice, it is essential to disaggregate the data lifecycle in GenAI development, since each stage introduces distinct risks and governance challenges. Foundation model pretraining data is frequently proprietary or undisclosed by vendors, making full lineage assessment infeasible [74]. This opacity necessitates focus on due diligence on later stages of adaptation and alignment, where data choices are under direct control.

Specifically, data used for supervised fine-tuning and reinforcement learning from human feedback (RLHF) introduces targeted risks: fine-tuning corpora can encode spurious correlations or domain-specific biases [75, 67], while preference datasets in RLHF are often small and subject to annotator subjectivity [76]. Similarly, datasets used to filter or suppress undesirable behaviors (safety alignment, tone of voice adjustment, instruction-following) should themselves be assessed for coverage and bias. If such filter datasets under-represent harmful behaviors, residual vulnerabilities will persist in production [77].

For financial applications, internal data leveraged to improve reasoning, scenario analysis, or domain adaptation (For example, transaction histories, regulatory text, or client communications) should be subjected to stringent quality assessments. This includes checks for representativeness, recency, contamination by synthetic or vendor-generated content, and considerations towards compliance with data protection regulations [78]. Unlike opaque pretraining, these datasets fall squarely within the control of the developers and therefore define the boundaries of acceptable model behavior for regulated tasks.

Finally, each downstream task or deployment context requires a distinct data assessment. A generative model fine-tuned for producing analyst reports should be evaluated on corpora reflecting reporting standards and regulatory disclosure norms, whereas the same model adapted for customer communications should be tested against consumer protection and clarity-of-language criteria. The emphasis is therefore on *context-specific assessment pipelines*, where task relevance, coverage, and risk materiality are explicitly documented [79]. This layered approach recognizes that a single assessment cannot necessarily cover all stages of a composable GenAI workflow, and that each adaptation stage may alter the risk surface of the model.

3.3.1 Data Quality Risks in GenAI

Distribution shifts. Foundation models are trained on broad, multi-domain corpora whose distributions can diverge from deployment settings (domain, temporal, or geographic drift), yielding sharp performance decay [80]. Recent work also shows that feedback loops—such as retraining on model-generated content or crawls polluted by synthetic text—induce “model collapse” and semantic drift even at modest rates of synthetic contamination [14]. For financial applications, this implies explicit dataset time-stamping, recency controls, and out-of-distribution (OOD) detection for production inputs.

Spurious correlations. At web scale, high-capacity models readily exploit shortcuts unrelated to task semantics (stylistic cues, lexical artifacts), leading to brittle behavior under perturbation or domain shift [81, 82]. Because such correlations are deeply embedded during pretraining, institutions should expect distributionally robust evaluation and causal/stress tests (such as counterfactual re-writes) to reveal hidden reliance on non-causal features prior to use in high-stakes workflows.

Biased data. Pretraining corpora encode demographic, cultural, and socio-economic skew. These biases surface as disparities in generated content (tone, framing, stereotypes) and can be amplified post fine-tuning [83, 84]. Contemporary evaluation suites (For example, HELM) report fairness and toxicity axes alongside accuracy, underscoring the need to track group-conditional error and content safety during development.

Adversarial poisoning. Targeted poisoning at small fractions of large corpora has been shown to be *practically* feasible and effective against modern LMs [65, 85]. Poisoned samples can induce task-specific failures, embed false facts, or steer model behavior under trigger prompts. Development pipelines therefore require ingestion-time defenses (source whitelisting, reputation signals), robust filtering, and canarying to detect unexpected behavioral shifts across model snapshots.

Backdoor attacks. Backdoors—latent behaviors activated by rare triggers—have been demonstrated in NLP models and extended to instruction-tuned LLMs, where triggers can be token patterns or formatting artifacts [86, 87]. Because backdoors may remain dormant under conventional tests, firms should combine trigger search/steering probes with rule- and verifier-based audits and isolate contaminated shards during dataset assembly.

While Adversarial poisoning and backdoor attacks are strictly speaking issues relating to cyber security and information security, model validators would typically be well equipped to analyze and examine such propensities, or notice potential vulnerabilities.

In line with SR 11–7, these risks motivate *documented* controls: dataset documentation (datasheets), provenance and license tracking, deduplication and contamination scans, bias audits with group-conditioned metrics, time-based snapshotting, and change logs that tie dataset versions to model checkpoints. Where training or adaptation data are vendor-supplied, institutions should obtain attestations covering sources, filtering criteria, and known limitations, and treat undisclosed data lineage as a material model risk factor

[74].

Establishing relevance and sufficiency of data used during development requires that one carefully consider and track data quality at collection (or acquisition from a vendor) and throughout training. Avoiding serious pitfalls from bias, spurious correlations, or backdoor attacks requires planning and documented mitigations.

3.4 Uncertainty Quantification

All model outputs should be viewed as estimates and quantifying and managing the uncertainty of these estimates is particularly challenging for generative AI, where outputs are complex, open ended, and context dependent. In financial applications, failing to understand when a model is uncertain can introduce material risks. Supervisory guidance reflects this concern: SR 11-7 explicitly calls for accounting for “*model uncertainty*” through conservative use and adjustments [6], while SS1/23 requires firms to demonstrate that “*model uncertainties are adequately understood, monitored, and managed*” [8]. In practice, this means firms must be able to measure uncertainty reliably, justify their chosen methodology, and embed fallback mechanisms where uncertainty exceeds acceptable thresholds.

Taxonomy of Methods

Quantification of uncertainty in GenAI systems is largely an open problem and current approaches can be categorized into several dimensions. Beyond the classical split between *aleatoric uncertainty* (irreducible randomness in data) and *epistemic uncertainty* (stemming from limited knowledge) in model parameters and predictions, recent work highlights uncertainty along new axes such as input ambiguity, reasoning path divergence, and stochastic decoding variability [88]. Methods can also be grouped by their computational approach: single-pass heuristics, multi-pass sampling or ensembling, and post-hoc calibration.

Predictive Confidence from Model Internals

One of the most direct signals of uncertainty is the token- or sequence-level log-likelihood of outputs. Low probability assignments can serve as proxies for lack of confidence [89]. **Advantages:** straightforward to implement on models exposing token probabilities; aligns with SR 11-7’s principle of “knowing what you know.” **Limitations:** raw probabilities are often miscalibrated, leading to overconfidence, and require additional post-processing to be interpretable.

Prompt-Driven Epistemic Uncertainty

Iterative or varied prompting can reveal epistemic uncertainty by exposing disagreement in the model’s own responses [90, 91]. This is analogous to challenger models in traditional MRM, where the same problem is posed in multiple forms. **Advantages:** model agnostic; can be applied even in closed API settings. **Limitations:** computationally expensive; estimates may be unstable and sensitive to prompt design.

Semantic-Level Measures

Rather than relying solely on surface-form probabilities, semantic similarity across multiple outputs can quantify uncertainty at the meaning level. *Semantic entropy* [91] captures the dispersion of generated meanings, even when surface variation is low. **Advantages:** aligns more closely with user perception of reliability; robust to paraphrase variance. **Limitations:** requires auxiliary embedding models, adding complexity and potential secondary risk.

Ensemble and Diversity-Based Methods

Building on principles from classical model risk management [92], ensembles of independently trained or perturbed models provide uncertainty estimates via disagreement. More recent approaches such as MUSE [93] approximate ensembles efficiently by leveraging subset divergence. **Advantages:** empirically strong; naturally captures epistemic uncertainty. **Limitations:** high computational cost; often infeasible for closed-source GenAI APIs.

Calibration and Postprocessing

Post-hoc calibration techniques, such as temperature scaling, isotonic regression, or LLM-specific fine-tuning, adjust model confidence estimates to better align with empirical correctness rates [94, 95].

Advantages: low implementation cost; improves interpretability and regulatory defensibility. **Limitations:** depends on the availability of representative, high-quality validation datasets.

Conformal Prediction Approaches

Emerging methods apply conformal prediction to generative tasks, producing sets of plausible outputs with guaranteed error rates [96]. For instance, in classification-style tasks, conformal prediction can bound the probability that the true label lies outside the proposed set. Extending these techniques to open-ended text generation is an active area of research [?]. **Advantages:** provides formal statistical guarantees and aligns well with regulatory demands for measurable reliability. **Limitations:** adapting conformal prediction to multimodal, freeform outputs remains challenging.

Reasoning-Focused Uncertainty

For reasoning-heavy tasks, divergence in intermediate steps is itself an indicator of uncertainty. Techniques such as measuring entropy across multiple chain-of-thought paths or aggregating answers across self-consistency sampling highlight when reasoning is unstable [97]. These approaches are particularly relevant in finance, where errors in reasoning chains (such as multi-step scenario analysis) can have outsized consequences.

Implications for Risk Management

In practice, uncertainty quantification should combine several of these methods into a hybrid framework. Single-pass token log-likelihoods or calibrated confidence scores can provide lightweight screening, perturbation- and prompt-based disagreement can detect epistemic fragility, and conformal or ensemble approaches can provide stronger, if costlier, guarantees. For high-stakes financial workflows, UQ must be tightly coupled with governance: when thresholds are exceeded, models should trigger conservative adjustments (such as deferment to human review, disclaimers, or fallback models) [6, 8].

From a regulatory perspective, the objective is not to eliminate uncertainty but to demonstrate that it is systematically understood, documented, and controlled. Robust UQ practices thus act as a safety gauge, ensuring that GenAI systems are deployed with awareness of their limitations and within the boundaries of acceptable risk.

Proper uncertainty quantification is critical to calibrated risk estimation for GenAI models. While, uncertainty quantification for GenAI models remains an active area of research, best practices suggest that it uncertainty can be meaningfully quantified in ways that can reduce model risk.

4 Model Testing and Monitoring

4.1 Assessing Quality and Relevance of Test Data

Unlike traditional statistical models with hand-crafted features—which can often be grounded in established statistical or mathematical theory and thereby offer a degree of interpretability—Generative AI (GenAI) workflows are largely evaluated empirically. This shift has profound implications for model risk management. In classical credit or market risk models, one could directly connect features to theory (such as option pricing equations, logistic regression coefficients), providing transparent and auditable justification. By contrast, foundation models and argentic workflows rely on high-dimensional learned representations whose behavior cannot be traced back to first principles. Consequently, the **choice of test data and benchmarks becomes paramount**: it determines whether a model’s empirical evaluation is meaningful, reliable, and aligned with its intended use [98, 99, 100, 101]. As such, one should carefully inspect the datasets employed at development and understand when the datasets are appropriate and what assumptions they encode.

Moreover, it is particularly important to consider the need to purchase data from vendors or contractors as it is increasingly the case that advanced AI models are evaluated using human studies that are conducted in-house or commissioned. Technologies used include Amazon Mechanical Turk or Prolific, and should be well documented to ensure alignment with ethical principles of human studies and ensure the study was conducted in a manner that accurately reflects some aspect of the business use case. If engaging with a third party vendor for such activities, all comments in our section on vendor relations become relevant.

Testing Data Risks

Even if a business risk has been perfectly mapped to a set of quantifiable metrics, the utility of computed estimates for risk assessment is only as valid as the data used. If unrealistic scenarios are considered or even if realistic scenarios are considered with misleading probabilities, testing efforts will lead to miscalibrated risk estimates. Within GenAI applications, evaluation datasets (often termed ‘evals’) are continuously emerging and are widely viewed as best practice for systematic assessments of various characteristics of AI models including performance, accuracy, biases, and safety using benchmarks and testing frameworks. Public leaderboards and benchmark suites using the existing benchmarks can serve as a valuable baseline to understand if model performance is within standard limits. New benchmarks and standardized tests and metrics have been designed to evaluate and compare various categories of risk for general-purpose AI applications in stylized scenarios and tasks [102, 103, 104, 105, 106]; however, choosing the correct benchmark and evaluation to suite the stated business use case is paramount for ensuring that the model performs as intended. Generally, benchmarks can be broken down by their characteristics:

Domain-specific benchmarks are focused on assessing a model’s performance on a particular task that requires domain-specific knowledge. For example, GSM8K [107] and MATH [102] are datasets of mathematical problems at grade-school and high-school level respectively, while SWE-Bench [103] tests software engineering problems from real GitHub issues and Python repositories.

General capability benchmarks evaluate the fitness of an LLM in a general sense, applying the model to a wide range of inputs or tasks and considering performance under a certain theme. For example, HellaSwag [108] provides the model with a context and a multiple-choice question for which multiple answers given are plausible but only one is a logical response to the prompt according to ‘common sense’. The evaluation therefore tests the ability of the model to differentiate logically and contextually consistent text from facially relevant but nonsensical text.

Safety benchmarks aim to evaluate the safety, robustness, and alignment of AI models, ensuring they behave as intended and avoid harmful outcomes. For example, RealToxicityPrompts [44] tests AI models for harmful or toxic language generation by providing prompts and measuring the likelihood of unsafe outputs.

Moreover, while the existing benchmarks above may likely not be linked to concrete business risks, reading their source documentation to understand the considerations taken during their construction can aid in constructing datasets to evaluate claims if necessary. We discuss further prominent examples of such datasets below.

4.1.1 Domain-Specific Benchmarks

Domain-specific benchmarks are designed to evaluate model performance within specialized application areas such as healthcare, finance, and law. They provide task-relevant datasets that reflect the requirements, risks, and regulatory sensitivities of these domains. For example, in law, **LegalBench** evaluates large language models on statutory interpretation, legal reasoning, and case analysis tasks [109]. In finance, datasets such as **FinQA** [110], **ConvFinQA** [111], and **TAT-QA** [112] focus on numerical reasoning across textual and tabular data, making them directly relevant for banking and regulatory reporting use cases.

These benchmarks are increasingly combined with general reasoning datasets such as **MMLU** [113] and **GPQA** [104], which extend evaluation to complex, expert-level knowledge. While domain-specific datasets ensure task alignment, empirical studies show that general-purpose reasoning tests highlight cross-domain transferability and robustness under new conditions. Together, these benchmarks provide complementary insights: domain sets measure applicability to specific business contexts, while broader tasks stress-test adaptability and logical reasoning capabilities. In practice, financial institutions adopting GenAI workflows should integrate both types of evaluation to demonstrate that models work as intended and sufficiently robust for use.

4.1.2 General-capability benchmarks

General-capability benchmarks assess a model's overall performance across a broad range of tasks, measuring its reasoning, comprehension, adaptability, and problem-solving abilities. Unlike domain-specific benchmarks, which focus on specialized knowledge, general-capability benchmarks evaluate foundational AI skills such as logical reasoning, common-sense understanding, multilingual proficiency, and problem solving.

Several specialized benchmarks further refine general capability evaluation. **MUSR** [114] assesses multi-step reasoning through machine-generated narratives with complex questions in domains like murder mysteries and object placement, validating the entire reasoning tree. **IFEval** [115] measures a model's ability to follow explicit instructions by checking adherence to specific directives, such as mentioning a keyword multiple times. **BBH (BigBenchHard)** [116] selects the most challenging tasks from the BigBench dataset, focusing on those with objective validation metrics that even human testers find difficult. Additionally, the **Cohere Multilingual Capability** benchmark suite evaluates language models' proficiency across multiple languages, ensuring robust multilingual performance. These benchmarks enhance the assessment of general intelligence, logical consistency, and adaptability in AI systems.

Finally, existing ML competitions help researchers and practitioners to capture the current evaluation interests of the community [117].

4.1.3 Safety benchmarks

General use of conversational AI in a bank, either as a customer facing application, or internally facing application are much more specific and general publicly available ChatBots. Hence, banks would not likely expose a ChatBot to its customers or employees, without strict control of both input and output, ensuring it is both specific and appropriate.

Hence, safety benchmarks related to general use of less constrained conversational AI are less relevant in a banking setting but included here mostly for completeness. Also, issues connected to safety, legality, intellectual property, information security (or even business risk such as Return on Investment), is not

classically considered within the scope of model risk management. However, for certain applications of GenAI, these aspects might still be relevant.

Safety benchmarks assess risks related to harmful outputs, ethical concerns, robustness against adversarial inputs, and compliance with safety guidelines. They help ensure models behave responsibly, avoid generating misinformation, and adhere to ethical AI principles.

In our categorization, we first assess trustworthiness characteristics as defined by the NIST RMF and classify safety benchmarks into four key areas: security, privacy, fairness, and overall safety. Some benchmarks encompass multiple aspects of safety, providing a holistic evaluation of AI risks. For instance, the **MLCommons AI Safety Benchmark** [118] measures the safety of LLMS by evaluating responses to prompts across multiple hazard categories, such as child sexual exploitation, indiscriminate weapons, and self-harm risks.

The Sociotechnical Safety Evaluation Repository [119] offers a collection of benchmarks and evaluation methods to help developers assess societal risks associated with LLMS and other generative AI systems. Additionally, **Harmful Capability Evaluations** [120] focus on determining whether general-purpose AI possesses particularly dangerous knowledge or skills.

It is paramount to ensure the data used to evaluate and test models is conceptually sound and accurately represents the business use case of the developed model. Data can be sourced via commission or publicly available benchmarks with specific intents can be employed including domain-specific, general capability, and safety benchmarks.

4.2 Understanding Coverage of Edge-Case Risk with Stress Testing

Understanding edge-cases and rare-events is of paramount importance to properly calibrated model risk. Given the expressivity of modern neural networks (the foundational technology used across GenAI), the ability for minor—often imperceptible—modifications to inputs, model parameters, or distributions to cause catastrophic shifts in model behavior must be considered.

We highlight that in this section we do not discuss challenger models as baselines as covered by SR 15-18, but focus on requirements set out by SR 11-7:

Where appropriate to the particular model banks should employ sensitivity analysis in model development and validation to check the impact of small changes in inputs and parameter values on model outputs to make sure they fall within an expected range.

Naturally, ensuring that development tests have selected the correct level of stress-testing always depends on the model context and materiality.

4.2.1 Stress testing

One approach to overcoming the static nature of evaluation benchmarks is to assume data can be dynamically altered to minimize the model performance with respect to a particular metric. In a business sense, this is a kind of worst-case analysis; within machine learning this is often discussed as *adversarial* quantification as it assumes some adversary wants to minimize model performance. Though not a realistic business abstraction, it is a helpful abstraction that unifies relevant emerging research. The parameters of change to a model or dataset is generally termed the *threat model* and encompasses (1) what information that can change or that outsiders have access to and (2) the parameters of the change that can reasonably be made to a system. The primary benefit of adversarial stress testing is that it is a general framework that transforms static benchmarks into dynamic benchmarks by introducing assumptions about the types of dataset shifts/changes one may see at deployment time. Adversarial stress testing can be performed at different levels of *rigor* from random perturbations to formal mathematical guarantees of performance. We highlight different levels of adversarial stress testing below:

Statistical adversaries Statistical adversaries aim to degrade model performance by exploiting naturally occurring statistical variations in the data. Threat models can be as simple as assuming the same distributions one might commonly use for dataset augmentation [121]. The critical difference between this and augmentation is the use of augmentation to search for a *worst-case* input that minimizes a metric value [122, 123]. These methods are particularly useful for understanding a model's sensitivity to data shift, out-of-distribution samples, or underrepresented edge cases [124, 125, 126]. Such attacks are relatively uncommon on generative AI systems, though the straightforward extension of the works cited here (which is no doubt in operation) is to model a distribution of malicious prompts, prefixes, and suffixes in the case of natural language and to test the system against this distribution. For example, *AdvBench* in [127] catalogues hundreds of examples of malicious examples.

Falsification adversaries Falsification adversaries attempt to identify concrete failure cases in a model by systematically searching for inputs that lead to incorrect or undesirable outputs. These inputs are popularly known as adversarial examples [128, 129, 130]. Unlike statistical adversaries, which rely on probabilistic perturbations, falsification adversaries use structured search techniques [131, 132], such as genetic algorithms [133] or optimization-based methods [130], to maximize the probability of failure [134]. This approach is widely applied in safety-critical domains, such as autonomous systems and finance, where failure cases must be explicitly catalogued and mitigated. This has been extended to both text classification in the case of the *HotFlip* attack [135] and to aligned language models [127].

Certification against adversaries Certification adversaries operate within a mathematically rigorous framework, providing formal guarantees on model robustness against predefined classes of adversarial attacks [136, 137, 138]. These techniques use worst-case optimization [138], formal verification [136], and statistical techniques [139] to establish provable performance bounds under adversarial conditions. While computationally expensive, certification adversaries provide the strongest assurances of model reliability, making them crucial for applications where failure has severe consequences, such as automated trading or medical AI systems. To the best of our knowledge the only such formal, deterministic certification of language models comes from [140] where the authors were able to certify the differential privacy of a fine-tuned GPT-2 model.

Several different approaches have been developed depending on the metric to be attacked. While not all metrics or business use cases will need to consider trustworthiness, one should be aware of the many failure modes that can be tested for and assess if they represent material risk. In what follows we keep with the taxonomy of trustworthy characteristics used by the machine learning community, but stress that these sorts of considerations should be made only when they are tied to a direct business risk.

Robustness Perhaps the most well established application of adversarial stress testing is in evaluating robustness [141, 142]. Early work in this area focused on generating adversarial examples—inputs subtly modified to deceive models while remaining indistinguishable to humans [128]. More recent approaches extend these methods to structured data [143], time-series forecasting [144], and reinforcement learning environments [145, 146, 147]. The goal is to identify worst-case scenarios where small input changes lead to disproportionate model failures. Attacks on language models have shown substantial success even with aligned, deployed models such as ChatGPT [127].

Explainability Explainability adversarial stress testing explores how model interpretations can be manipulated or degraded under adversarial conditions [148, 149]. For instance, feature attribution methods such as SHAP and LIME can be tested against adversarial perturbations to assess whether explanations remain stable when small, seemingly irrelevant modifications are introduced to the input [150]. Ensuring explanation consistency is critical for building trust in AI models used for high-stakes decision-making. Provided that generative models are often pretrained on data scraped from the internet and not housed on a secure centralized server, several attacks have demonstrated that this is a practical threat vector against GenAI [65].

Privacy In privacy-focused adversarial stress testing, the goal is to evaluate a model's susceptibility to data leakage [151] and membership inference attacks [152]. These tests help determine whether

private or proprietary information can be extracted from the model through repeated querying or gradient-based attacks [153]. As with other properties, these attacks have been generalized to the GenAI pipeline in recent years [154]. Techniques such as differential privacy and secure multi-party computation are often integrated into adversarial testing frameworks to assess their effectiveness against potential breaches. Recently, approaches for adversarial robustness have been adopted to aid in privacy analysis [140]. Data privacy is not classically a model risk concern, but have been included here mostly for completeness. However, as with some types of information security concerns, following model risk management practices might uncover issues related to data privacy. Training or evaluation datasets sometimes pose a special "privacy concentration risk", and models can contain representations of sensitive information that can potentially be extracted or exploited.

Fairness Fairness is not typically stress-tested but quantified [155]. Within algorithmic fairness, adversarial stress testing investigates how model predictions change across different demographic groups when exposed to adversarially manipulated inputs [156, 157]. This helps uncover cases where a model is unfairly biased against certain subpopulations [158], either due to dataset imbalances or implicit algorithmic biases. Adversarial techniques can be used to generate counterfactual scenarios that reveal disparities in model decision-making, guiding the development of fairness-aware learning techniques [159, 156]. One substantial issue that traditional fairness methods face in GenAI pipelines is the fact that generative AI often propagates *representational harm* and not *allocative harm*. That is, GenAI does not deprive subgroups of particular resources but instead reinforces subordination of subpopulations via the propagation of harmful stereotypes [83]. This can be found in GenAI on text [160] and images [161].

Limitations and challenges: Adversarial stress testing, while valuable for evaluating model robustness, security, and fairness, faces several key challenges. One major limitation is the computational complexity of generating adversarial examples, particularly for certification-based approaches that require solving difficult optimization or verification problems. Additionally, the effectiveness of stress testing depends on the assumed threat model—if too weak, meaningful vulnerabilities may be missed; if too strong, the generated adversarial scenarios may not reflect realistic deployment conditions. Moreover, adversarial stress tests often lack clear generalization to real-world data shifts, making it difficult to assess their practical relevance. Additionally, adversarial threats evolve rapidly, often outpacing the development of robust defense mechanisms, leaving models vulnerable despite ongoing stress testing efforts.

4.2.2 Red teaming

Red teaming refers to adversarial interaction with an AI model, in order to elicit unwanted behavior or violate safety principles. While adversarial evaluation is strictly automated, red teaming can be fully manual (but is often a hybrid of manual efforts and automatic exploration). In particular, red teamers can take on the role of those seeking to misuse a model to design custom inputs that identify worst-case behaviors, malicious use opportunities, and unexpected failures.

While red teaming is classically an information security discipline, and does not necessarily relate to model risk issues, several of the approaches used in red teaming can be used to probe the robustness and edge cases of model behavior, that is relevant from a model risk management perspective.

Importantly, unlike adversarial tests, red teaming is able to consider nuanced interactions between models and external systems (such as API calls or IT infrastructure). In this sense, red teaming introduces a more system-level, business-specific dynamic to validation activities. In this section we review some of the kinds of activities that red teams often undertake.

'Jailbreaking' attacks are a common class of attacks used in red teaming, aiming to disable the models' safety restrictions [162, 127, 163, 164, 165, 166]. Attacks against language models can take the form of automatically generated inputs [127, 164, 167, 168, 77, 169, 170, 171, 172, 173] or manually generated ones [174, 175, 169, 176].

Red teaming practices in industry diverge along several key axes, including the setting (such as the characteristics of red teamers and the resources and methods available to them), and the decisions

it informs (For example, subsequent reporting, disclosure, and mitigation) [177]. Benchmarks can be used for exploratory purposes to craft a set of problems to induce a differential response that is against policy—for example, a set of prompts that vary only by the race of a mentioned subject can identify bias or discriminatory behavior. However, manual red teaming over longer conversations can catch issues that current automated attacks alone can miss [174].

Limitations and challenges Despite its effectiveness, red teaming faces several challenges that limit its scalability and impact. Manual red teaming can uncover complex failure modes that automated attacks may miss, but it is slow, resource intensive, and requires specialized expertise and access. Additionally, existing automated red teaming techniques raises some concerns over their 'ecological validity', whether remaining limited in scope and ability to be generalized to real-world scenarios [178]. Further research for faster and effective automated red-teaming is necessary to address this challenge [179].

Another major challenge is the reactive nature of red teaming—it identifies vulnerabilities post-deployment rather than proactively preventing them. Furthermore, benchmark-driven approaches may lead to narrow evaluations that fail to capture real-world adversarial risks.

Lastly, effective red teaming strategies must account for evolving threats, particularly as adversaries refine their methods to bypass safeguards. Addressing these challenges will require advancements in scalable automation, context-aware attack generation, and into broader risk management frameworks.

Modeling realistic deployment conditions and justification of model selection decisions can be conducted via stress-testing and potentially probed using red teaming approaches.

4.3 Model Monitoring

Similar to how static benchmarks, absent stress-testing, are generally not reflective of real-world deployment conditions, pre-deployment testing is typically insufficient for quantifying and mitigating model risk.

Ongoing monitoring is essential to evaluate whether changes in products, exposures, activities, clients, or market conditions necessitate adjustment, redevelopment, or replacement of the model and to verify that any extension of the model beyond its original scope is valid.

Traditional model risk management practices, developed around models such as logistic regression for credit scoring, would typically have dictated the following: (1) the establishment of a clear set of performance metrics (For example, Gini, AUC, KS-statistic); (2) monitoring of population stability via PSI (Population Stability Index) and CSI (Characteristic Stability Index); and (3) threshold-based alerts when drift or degradation in performance was detected. Remediation actions could then range from retraining to full redevelopment, depending on the severity of performance deterioration.

For GenAI systems, however, these methods may be insufficient as risks cannot be captured solely by scalar performance metrics, and failure modes often emerge in long-tail or interactional behaviors rather than averaged-out aggregate metrics. Thus, model monitoring requires extending the principles from traditional models to high-dimensional, generative, and dynamic systems.

Pitfalls with Model Monitoring

For GenAI and agentic systems, monitoring presents distinctive challenges:

- **Hidden failure modes in long-horizon use:** Agentic systems may perform well on single-turn or short tasks but degrade in multi-step workflows. Failures often emerge only after extended interaction or task chaining, making them invisible to standard monitoring dashboards.
- **Exposure–reliance mismatch:** An agent may be lightly used overall but deeply embedded in a critical workflow (high reliance), or conversely, widely used for experimentation but in low-stakes

settings (high exposure). Without explicitly separating these two dimensions, monitoring frameworks risk over- or under-weighting risks.

- **Monitoring blind spots:** Many GenAI failures are qualitative and context dependent, making them hard to capture in scalar metrics. This creates blind spots where risk accumulates undetected until an incident occurs.

4.3.1 Futureproofing Model Performance

While monitoring has traditionally been retrospective—tracking accuracy, stability, and error rates against historical baselines—the complexity of GenAI workflows requires a more forward-looking approach. Futureproofing model performance entails anticipating how systems may behave under evolving data distributions, infrastructure updates, and organizational use cases, ensuring that risks are identified and mitigated before material impacts occur.

Temporal Monitoring Pipelines. Continuous logging and temporal segmentation of model performance enable risk managers to analyze pre- and post-update behavior. This is particularly important when workflows undergo changes such as API upgrades, prompt-template modifications, or reindexing in retrieval-augmented generation (RAG) systems, where the model's behavior may shift abruptly despite no changes in core weights. Temporal drift analysis provides the empirical basis to separate natural distribution shifts from system-induced behavioral changes [180].

Drift and Stability Detection. Futureproof monitoring incorporates statistical drift detection techniques to flag deviations from baseline distributions. Methods such as population stability indices, kernel-based two-sample tests, and embedding similarity measures have been adapted to large-scale LLM monitoring [181]. In financial contexts, where small deviations can compound into material risk, drift alerts allow early escalation to human reviewers or automated fallbacks.

Scenario-Based Nowcasting. Finally, historical monitoring data can be leveraged for predictive “now-casting”—using time-series of model behaviors to anticipate future degradation or failure modes. By mapping behavioral trajectories across updates, institutions can proactively simulate whether planned changes (such as fine-tuning with new data) are likely to introduce bias, hallucinations, or regulatory non-compliance. This practice aligns with supervisory principles of proportional and ongoing oversight under SR 11-7 and SS1/23, extending monitoring from a retrospective safeguard into a predictive control mechanism.

Established Monitoring Practices

Several monitoring strategies have been developed to address the unique risks of GenAI and agentic systems:

- **Prompt and response drift detection:** Beyond monitoring model inputs for statistical drift, GenAI systems can be tracked in embedding space or through clustering of prompts/responses, surfacing shifts in how users (or adversaries) interact with the model.
- **Event logging:** Continuous tracking, content filter activations, and unsafe response prevalence provides leading indicators of alignment failures. For agentic models, logging should also include tool-call misuse and failure-to-completion rates.
- **Scaled monitoring:** Monitoring intensity can be scaled according to both how widely the model is deployed across contexts and the materiality of the decisions it influences. For example, lightweight monitoring may suffice for creative ideation, but not for autonomous trade execution or tasks pertaining to regulatory compliance.
- **Adversarial probing pipelines:** Automated adversarial test suites (For example, jailbreak generators, prompt injection attacks, data exfiltration attempts) can be run in shadow mode to provide a dynamic benchmark of model resilience over time.

- **Human-in-the-loop audits:** Regular expert review of sampled interactions, including multi-turn dialogues and agent-executed plans, helps identify nuanced harms such as representational bias, covert stereotypes, or subtle safety bypasses that automated checks cannot capture.
- **Behavioral anomaly detection:** For agentic systems, monitoring task plans, external API calls, or resource usage patterns for anomalies provides an additional layer of defense against unintended or malicious behaviors.

Model Deprovisioning as Mitigation

Monitoring frameworks must also define clear triggers for *model deprovisioning*—the systematic rollback, restriction, or retirement of a model when monitoring indicates persistent or material risk. Deprovisioning depends on the particular use and is conditional on specific quantification of risk and established risk limits and risk appetite. Deprovisioning should be framed not as failure, but as an essential component of responsible lifecycle management.

Ensuring that data quality risks such as distributional drift, adversarial misuse, and representational harm do not cause substantial issues during model validation requires proactive monitoring. Effective mitigation should include automated drift detection and alerting pipelines, human-in-the-loop escalation channels for nuanced risks, and clearly defined deprovisioning protocols that allow safe rollback when model behavior persistently violates the risk tolerances set out in early stages of development.

5 Concluding Remarks

GenAI is playing an increasingly prevalent role in decision-making, risk assessment, and operational processes across financial institutions, owing to their performance advantages over traditional modelling approaches. Further, substantial increases in the scope and prevalence of such systems are forecasted across the financial services industry. While GenAI offers the potential for substantial improvements across industry functions, like all models, they introduce model risks that need to be understood and managed. In this paper, we have discussed important areas for established model risk management teams to consider when developing, deploying and validating models. In particular, we discussed the need to carefully consider model risk classification and tiering, thoughtfully evaluate and document design decisions during development, and to establish robust and repeatable testing procedures that can serve as the basis for monitoring approaches. As with the deployment of any new model, specific applications of GenAI will no doubt bring novel, concrete technical challenges. Through comprehensive model risk management practice these challenges can be systematically overcome to ensure deployed models are within risk tolerance, to reap the benefits of powerful new model techniques and applications, while safeguarding the integrity and stability of the financial system.

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